

SrTiO₃

$Pm\bar{3}m$ · cubic perovskite
Goldschmidt $t = 0.95$

GdFeO₃

$Pnma$ · distorted perovskite
Goldschmidt $t = 0.80$

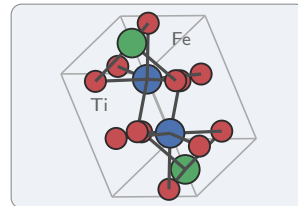
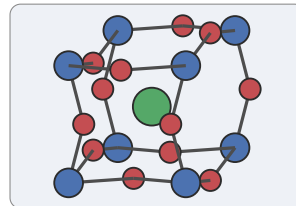
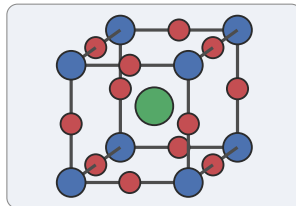
FeTiO₃

$R\bar{3}$ · ilmenite
Goldschmidt $t = 0.70$

(a) Structure

unit cell, schematic

- A-site cation
- B-site cation
- X-site anion

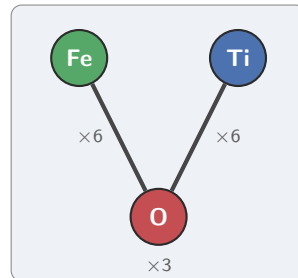
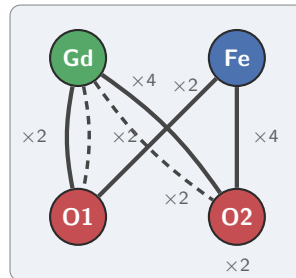
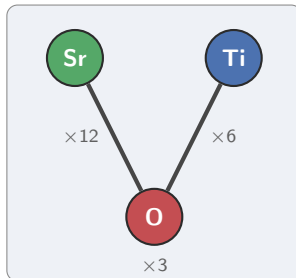


(b) “Bonding” edges

— core edge

- - - extended edge

$\times N$: bonds of that class per cation (edge); sites per formula unit (node)



(c) Polyhedral edges

— corner sharing

— edge sharing

— face sharing

— 4+ sharing

loop: sharing between symmetry-equivalent sites

